Fabio Gironella

Curriculum Vitae

Positions

Since 10/2022 **Chargé de recherche CNRS**, *Laboratoire Jean Leray*, Université de Nantes.

09/2020–09/2022 **Post-doctoral researcher**, *Humboldt University*, Berlin.

09/2018–08/2020 **Post-doctoral researcher**, *Alfréd Rényi Institute of Mathematics*, Budapest.

Education

07/2018 **Ph.D. in Mathematics**, *École polytechnique*, Palaiseau.

06/2015 **Diploma**, *Ecole Normale Supérieure*, Paris.

09/2014 **Master's Degree in mathematics**, *Université Pierre et Marie Curie*, Paris.

06/2012 **Laurea Triennale (Bachelor)**, *Università di Pisa and Scuola Normale Superiore*.

(Pre)prints

1. **On some rigidity results for high-dimensional confoliations**, *with Seungook Yu*.
   arxiv:2610.13927.
2. **C^0-Poisson geometry, coisotropic submanifolds, and clean intersection points**, *with Robert Cardona*.
   arxiv:2602.20388. Submitted.
3. **Vanishing cycles of symplectic foliations**, *with Klaus Niederkrüger and Lauran Toussaint*.
   arXiv:2410.07090. Submitted.
4. **Fillability obstructions for high-dimensional confoliations**, *with Robert Cardona*.
   arXiv:2410.06918. Submitted.
5. **Exact orbifold fillings of contact manifolds**, *with Zhengyi Zhou*.
   To appear in *Journal of Topology*. Also available at arXiv:2108.12247.
6. **Operator-Projected Variational Quantum Imaginary Time Evolution**, *with Aeishah Ameera Anuar, Francois Jamet, Fedor Simkovic IV and Riccardo Rossi*.
   J. Phys. A: Math. Theor. 59 245301 (2026). DOI 10.1088/1751-8121/ae767d. Also available at arXiv:2409.12018.
7. **Tight contact structures without symplectic fillings are everywhere**, *with Jonathan Bowden, Agustin Moreno and Zhengyi Zhou*.
   J. Eur. Math. Soc. (2026), published online first. DOI 10.4171/JEMS/1787. Also available at arXiv:2211.03680. Featured in this article of Quanta Magazine and in this episode of the podcast Co.Scienza. Submitted.
8. **Non-density results in high dimensional stable Hamiltonian topology**, *with Robert Cardona*.
   J. Lond. Math. Soc. (2) 111 (2025), no. 4. DOI:10.1112/jlms.70143. arXiv:2407.01357.
9. **Examples of symplectic non-leaves**, *with Lauran Toussaint*.
   Rev. Mat. Iberoam. 41 (2025), no. 3, 1081–1100. DOI:10.4171/RMI/1520. arXiv:2106.02618.
10. **Existence of conformal symplectic foliations on closed manifolds**, *with Lauran Toussaint*.
    Mathematische Annalen, 390, 2281–2335 (2024). DOI:10.1007/s00208-023-02729-0. arXiv:2111.00550.
11. **Asymptotically holomorphic theory for symplectic orbifolds**, *with Vicente Muñoz and Zhengyi Zhou*.
    Mathematische Zeitschrift 304, 21 (2023). DOI:10.1007/s00209-023-03277-8. arXiv:2201.09362.
12. **Bourgeois contact structures: tightness, fillability and applications**, *with Jonathan Bowden and Agustin Moreno*.
    Invent. Math. 230, 713–765 (2022). DOI:10.1007/s00222-022-01131-y. arXiv:1908.05749.

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13. **Examples of contact mapping classes of infinite order in all dimensions**.
Mathematical Research Letters, vol. 23, no. 3, pp. 707–727. DOI:10.4310/MRL.2021.v28.n3.a5. arXiv:1809.07762
14. **A remark on the contactomorphism group of overtwisted contact spheres**, *with Eduardo Fernández*.
Comptes Rendus Mathématique, 358 (2020) no. 2, pp. 189-196. DOI:10.5802/crmath.32. arXiv:1910.01359.
15. **On some examples and constructions of contact manifolds**.
Mathematische Annalen, 376(3), 957-1008. DOI:10.1007/s00208-019-01895-4. arXiv:1711.09800
16. **Examples of non-trivial contact mapping classes for overtwisted contact manifolds in all dimensions**.
Algebr. Geom. Topol., 19, n. 3 (2019), 1207-1227. DOI:10.2140/agt.2019.19.1207. arXiv:1704.07278

Theses

07/2018 **On some constuctions of contact manifolds**, *PhD thesis*, École polytechnique, Palaiseau. Under the supervision of Patrick Massot.

09/2014 **Vogel's theorem: uniqueness of contact structures approximating a foliation**, *Master's project*, Université Pierre et Marie Curie, Paris. Under the supervision of Patrick Massot.

06/2013 **An introduction to foliations on manifolds**, *Magister's project (joint work with Gabriele Rembado)*, Ecole Normale Supérieure, Paris. Under the supervision of Claude Viterbo.

06/2012 **Measured foliations on surfaces and classification theorems**, *Bachelor's project*, Università di Pisa, Pisa. Under the supervision of Bruno Martelli.

Conference/workshop/school invited talks

08/2026 **Workshop on symplectic topology 2026**, Belgrade University, Serbia.

10/2025 **School in symplectic and contact topology**, *University of Dakar, Senegal*.

08/2025 **Conformal dynamics and geometry in Bordeaux**, *University of Bordeaux, France*.

08/2023 **Workshop on Symplectic Field Theory 10**, *Humboldt University, Germany*.

07/2022 **Topologie**, *Oberwolfach, Germany*, (couldn't participate for health issues).

02/2020 **CAST 2020: XIV Workshop on Symplectic Geometry, Contact Geometry, and Interactions**, *University of Antwerpen, Belgium*.

01/2019 **Workshop on Symplectic and contact mapping class groups (two talks)**, *Seoul National University, South Korea*.

Seminar research talks

05/2026 **Séminaire Nord-Bassin Parisien "Structures géométriques de la dissipation"**, Online.

02/2026 **Arbeitsgemeinschaft Symplektische Topologie**, *Köln University*, Germany.

02/2026 **Poisson geometry research seminar**, *Köln University*, Germany.

01/2026 **Berlin-Hamburg-Hannover symplectic seminar**, *Hannover University*, Germany.

12/2025 **Séminaire de Systèmes dynamiques, Analyse et Géométrie**, *Avignon Université*, France.

12/2025 **Symplectic geometry seminar**, *ETH Zurich*, Switzerland.

09/2025 **Dutch Differential Topology & Geometry seminar**, *Vrije Universiteit Amsterdam*, Netherlands.

10/2024 **Topology seminar**, *Alfred Renyi Institute*, Budapest, Hungary.

05/2024 **Topology and Geometry seminar**, *Morningside Center of Math.*, Beijing, China.

01/2024 **Séminaire Symplectix**, *Institut Henri Poincaré*, Paris, France.

11/2023 **Topology seminar**, *Universitat de Barcelona*, Spain.

11/2023 **Differential geometry seminar**, *University of Minnesota*, Minneapoli, USA..

11/2022 **Séminaire Topologie et Géométrie**, *Université de Lyon I*, France.

10/2022 **Séminaire Nantes-Orsay**, *Université de Paris-Saclay*, France.

10/2022 **Séminaire Topologie, Géométrie et Algèbre**, *Université de Nantes*, France.

07/2022 **Rencontre symplectique Lyon-Grenoble (two talks)**, *Université de Lyon I*, France.

05/2022 **Oberseminar Geometrie**, *Universität München*, Germany.
03/2022 **Geometry and topology seminar**, *University of Uppsala*, Sweden.
11/2021 **Symplectic zoominar**, *CRM-Montréal, Princeton/IAS, Tel Aviv, and Paris*.
11/2021 **Seminar Symplektische Geometrie**, *Humboldt-Universität*, Berlin, Germany.
09/2021 **Geometry and Topology seminar**, *University of Toronto Scarborough*, Canada.
12/2020 **Seminar Symplektische Geometrie**, *Humboldt-Universität*, Berlin, Germany.
11/2020 **Floris Takens Seminar**, *University of Groningen*, Netherlands.
03/2020 **Topology seminar**, *Alfred Renyi Institute*, Budapest, Hungary.
01/2020 **Geometry and Topology seminar**, *UCLA*, Los Angeles, California.
11/2019 **Séminaire Nantes–Orsay**, *Université Paris Sud*, Orsay, France.
09/2019 **Monash Topology Seminar**, *Monash University*, Clayton, Australia.
07/2019 **Symplectic seminar**, *Humboldt-Universität*, Berlin, Germany.
04/2019 **Topology Wednesday meetings**, *Alfréd Rényi Institute*, Budapest, Hungary.
12/2018 **Geometry Seminar**, *Justus-Liebig-Universität*, Gießen, Germany.
12/2018 **Oberseminar Differentialgeometrie**, *Universität Augsburg*, Germany.
09/2018 **Geometry Seminar**, *Instituto de Ciencias Matemáticas*, Madrid, Spain.
03/2018 **Séminaire Géométries**, *Université de Lyon I*, France.
02/2018 **Séminaire de Topologie**, *Université Grenoble Alpes*, France.
11/2017 **Arbeitsgemeinschaft Symplektische Topologie**, *Universität zu Köln*, Germany.
10/2017 **Séminaire Géométrie Topologie Dynamique**, *Université Paris-Saclay*, France.
05/2017 **Oberseminar Geometrie**, *Universität München*, Germany.
05/2017 **Séminaire Symplectix**, *Institut Henri Poincaré*, Paris, France.

Hosted researchers

05/2026 **Shuo Zhang**, *Chinese Academy of Science*, 14 days.
03/2026 **Baptiste Serraille**, *ETH Zurich*, 6 days.
11/2025 **Wei Zhou**, *Universidad Complutense Madrid*, 7 days.
11/2025 **Pacôme Van Overschelde**, *Université Libre Bruxelles*, 5 days.
06/2025 **Vera Vertesi**, *University of Vienna*, 4 days.
04/2025 **Shuo Zhang**, *Chinese Academy of Science*, 14 days.
01/2025 **Thomas Massoni**, *Stanford University*, 7 days.
01/2025 **Cheikh Khoule**, *University Cheikh Anta Diop, Dakar*, 7 days.
12/2024 **Robert Cardona**, *University of Barcelona*, 7 days.
11/2014 **Diego Santoro**, *University of Vienna*, 3 days.
05/2024 **Agustin Moreno**, *Heidelberg University*, 4 days.
03/2024 **Francesco Morabito**, *ETH, Zurich*, 7 days.
03/2024 **Russell Avdek**, *IMJ-PRG, Sorbonne Université*, 7 days.
02/2024 **Simon Allais**, *Strasbourg University*, 3 days.
09/2023 **Zhengyi Zhou**, *Chinese Academy of Science*, 7 days.
07/2023 **Luca Accornero**, *Vrije University*, 7 days.
07/2023 **Alvaro del Pino Gomez**, *Utrecht University*, 7 days.
07/2023 **Lauran Toussaint**, *Vrije Universiteit Amsterdam*, 6 days.
07/2023 **Eduardo Fernandez**, *University of Georgia Athens*, 7 days.
03/2023 **Yusuke Kawamoto**, *ETH, Zurich*, 7 days.
11/2022 **Lauran Toussaint**, *Vrije Universiteit Amsterdam*, 6 days.

Research visits

06/2026 **Hannover University**, Germany, 5 days.
02/2026 **Universitat Koln**, Germany, 7 days.
12/2025 **Universitat de Barcelona**, Spain, 10 days.
09/2025 **Utrecht University**, Netherlands, 6 days.
06/2025 **Sorbonne Université, Paris**, France, 4 days.
05/2025 **Sorbonne Université, Paris**, France, 3 days.
11/2024 **Alfred Renyi Institute, Budapest**, Hungary, 6 days.
07/2024 **Université de Lyon I**, France, 6 days.
05/2024 **Morningside Center of Mathematics, Beijing**, China, 14 days.
11/2023 **Universitat de Barcelona**, Spain, 5 days.
04/2023 **ICMAT, Madrid**, Spain, 5 days.
03/2023 **Université Pierre et Marie Curie**, Paris, France, 3 days.
11/2022 **Université de Lyon I**, France, 6 days.
01/2022 **Université de Lyon I**, France, 6 days.
07/2021 **Universität Regensburg**, Germany, 5 days.
11/2019 **Mathematisches Forschungsinstitut Oberwolfach**, Germany, one week.
09/2019 **Monash University**, Clayton, Australia, two weeks.
07/2019 **Humboldt-Universität**, Berlin, Germany, 5 days.
06/2019 **Universität Augsburg**, Germany, one week.
01/2019 **Seoul National University**, Seoul, South Korea, one week.
01/2019 **MATRIX Mathematical Research Institute**, Creswick, Australia, 10 days.
12/2018 **Justus-Liebig-Universität**, Gießen, Germany, 3 days.
12/2018 **Universität Augsburg**, Germany, one week.
09/2018 **Instituto de Ciencias Matemáticas**, Madrid, Spain, one week.
05/2018 **Humboldt-Universität**, Berlin, Germany, 3 days.

Attended conferences/workshops

09/2026 **Workshop: Algorithms in Contact Topology**, University of Sevilla.
08/2026 **Symplectic topology - in honour of Kai Cieliebak**, Augsburg University.
08/2026 **Workshop on symplectic topology 2026**, Belgrade University.
08/2026 **Poisson conference 2026**, Leiden University.
06/2026 **Symplectic Topology, Hamiltonian Dynamics, and Persistence Structures. A conference in honour of Octav Cornea**, ETH Zurich.
02/2026 **Workshop on contact geometry, relativity and thermodynamics**, Stony Brook.
07/2024 **Symplectic Geometry and Anosov Flows**, University of Heidelberg.
06/2024 **From Hamiltonian Dynamics to Symplectic Topology and Beyond**, IHP, Paris.
04/2024 **Workshop on Higher-dimensional contact topology**, AIM, Los Angeles.
10/2023 **Rencontre ANR Cosy "Théorie de l'homotopie de Floer"**, IRMA, Strasbourg.
07/2023 **From smooth to C^0 symplectic geometry: topological aspects and dynamical implications**, CIRM, Marseille.
05/2023 **School on conformal symplectic dynamics and related fields**, CIRM, Marseille.
11/2022 **Floer Homotopical Methods in Low Dimensional and Symplectic Topology**, MSRI, Berkley (on Zoom).
07/2022 **Convexity in contact and symplectic topology**, IHP, Paris.
11/2021 **Locally conformal symplectic manifolds**, BIRS, Banff (on Zoom).

09/2021 **Topological Theory of Tangent Distributions**, Lorentz Center, Leiden (on Zoom).

04-06/2021 **Symplectic topology, contact topology, and interactions**, IHP, Paris (on Zoom).

01/2020 **Geometry and Topology workshop**, UCLA, Los Angeles.

12/2019 **Geometric methods in Symplectic Topology**, ICMAT, Madrid.

08/2019 **School and International Conference on Symplectic Topology**, IMPA, Rio de Janeiro.

06/2019 **Homological algebra, microlocal analysis and symplectic geometry**, Université de Montréal, Montréal.

05/2019 **Nemethi60 – Geometry and Topology of Singularities**, Alfréd Rényi Institute, Budapest.

03/2019 **XIII Workshop on Symplectic Geometry, Contact Geometry, and Interactions**, *CAST 2019*, Humboldt University, Berlin.

01/2019 **Topology of Manifolds: Interactions between high and low dimensions**, MATRIX Mathematical Research Institute, Creswick.

06/2018 **RIEMain in Contact**, *Conference/school in contact Riemannian geometry and contact topology*, Cittadella dei Musei, Cagliari.

08/2016 **Workshop on Symplectic Field Theory VIII**, Humboldt University, Berlin.

06/2016 **Developments in Contact and Symplectic Topology**, University of Glasgow, Glasgow.

04/2016 **Working group on microlocal invariants of Legendrian knots**, *ANR Microlocal*, CIRM, Marseille.

12/2015 **Flexibility and contact geometry**, Université libre de Bruxelles, Bruxelles.

09/2015 **Stein Manifolds, Contact Structures and Knots**, CIRM, Marseille.

06/2015 **Wrinkles and h-principles, old and new**, Ludwig-Maximilians-Universität München, Munich.

01/2015 **IX Workshop on Symplectic Geometry, Contact Geometry and Interactions**, *CAST 2015*, École normale supérieure, Lyon.

12/2014 **Flexibility and contact geometry**, Université libre de Bruxelles, Bruxelles.

07/2014 **Workshop on Contact Geometry in Dimension Three and Higher**, University College London, London.

Supervision

2026–2029 **PhD thesis (European Doctoral Network grant) of Lite Du**, Nantes Université.

2026–2027 **Postdoc (ANR Tremplin grant), Wei Zhou**, Nantes Université.

2024–2027 **PhD thesis of Maxime Moscatelli**, Nantes Université.

2024–2025 **Postdoc (Etoiles Montantes grant), Seungook Yu**, Nantes Université.

Spring 2025 **L3 research stage of Gabin Bourdeaux**, Nantes Université.

18–22 Nov 2024 **Group of five PhD students and Postdocs**, *collaboration with software firm TopSolid for the "Semaines d'Études Mathématiques – Entreprises et Société"* , Nantes Université.
One of the participating students has afterwards been hired by TopSolid.

Summer 2024 **Master 2 thesis of Maxime Moscatelli**, Nantes Université.

Spring 2024 **Master 1 thesis of Abdou Niang, Serigne Sarr and Maël Guillon**, Nantes Université.

Teaching

2025–2026 **Master course "Introduction to symplectic topology and fillability of contact manifolds"**, Nantes Université.

Oct 2025 **Mini-course on symplectic topology**, Dakar University.

2023–2024 **Master course "Introduction to symplectic topology via pseudo-holomorphic curves"**, Nantes Université.

2017–2018 **Exercise sessions in analysis**, École polytechnique, Palaiseau.

2016–2017 **Mentored the retake of the course *Introduction à l'analyse réelle***, École polytechnique, Palaiseau.

2016–2017 **Exercise sessions of *Formation préparatoire* for foreign students**, École polytechnique, Palaiseau.

2015–2016 **Supplementary exercise sessions for the course *Introduction à l'analyse réelle***, École polytechnique, Palaiseau.

Refereeing activities

1. **Referee** (quick opinions and detailed reports) for various **journals**, since 2020 till present. (More information available upon request.)
2. **Reviewer** for *MathSciNet*, since 2025.
3. **Reviewer** for *zbMATH* from 2020 till 2023.
4. **External refereee** for the **PhD thesis** of Jayden Hammett. (Supervisor: Diarmuid Crowley.) University of Melbourne, September 2026.
5. **Jury member** for the **PhD defense** of Wei Zhou (supervisors: Alvaro del Pino, Eduardo Fernandez). Universidad Complutense Madrid, January 2026.
6. **External refereee** for the **PhD thesis** of Dahyana Farias Uncovich (supervisors: Francisco Presas, Eduardo Fernandez), Universidad Complutense Madrid, January 2025.
7. **External referee** for the **PhD thesis** of Guillermo Sánchez Arellano (supervisors: Francisco Presas, Luis Giraldo Suarez). Universidad Complutense Madrid, April 2024.
8. **External refereee** for the **PhD thesis** and **jury member** for the **PhD defense** of Francisco Javier Martínez-Aguinaga (supervisors: Francisco Presas, Alvaro del Pino). Universidad Complutense Madrid, May 2023.

Organizational activities

1. **Co-organizer** a **school in contact and symplectic geometry** in Dakar, Senegal. Fall 2027. More information soon.
2. **Co-organizer** of **Workshop on Symplectic Field Theory XI** at Leibniz University, Hannover. August 2027.
3. **Co-organizer** of **school and conference on Contact Topology and Dynamics** at Nantes Université. May-June 2027.
4. Co-organizer of the **Colloquium** at Nantes Université, starting Sept 2026
5. Co-organizer of the math activities for science festival **Utopiales** in Nantes, Oct 2025
6. Co-organizer of **Séminaire Topologie, Géométrie et Algèbre** at Nantes Université, 2022–2025
7. **Co-organizer** of the *Workshop on locally conformal symplectic topology* at Nantes Université, 19-20 November 2025
8. **Co-organizer** of the *Spring school on singularities in low-dimensional and symplectic topology* at Nantes Université, 22-25 April 2025
9. **Co-organizer** of the *Summer school on Gromov-Witten invariants* at Nantes Université, 11-14 June 2023
10. **Co-organizer** of the math activities for the *Fête de la science* at Ecole Polytechnique from 2015 to 2018
11. **Co-organizer** of the *Tournoi Français des Jeunes Mathématiciennes et Mathématiciens* TFJM^2 from 2015 to 2017

Grants/Awards/Scholarships

05/2026 **Travel grant from CNRS**, *in the context of the Sino-French International Research Network*, 2000€.

10/2025 **ANR Tremplin – ERC Starting Grant (T-ERC StG) 2025**, *ANR grant for ERC StG 2025 candidates who ranked "A not funded" after interviews*, 115700€.

08/2025 **Marie Sklodowska-Curie Actions Doctoral Network (MSCA-DN) - TopSCoLDs**, *network of twelve supervisors for twelve PhD scholarships in total*, 3700000€.

01/2024 **Etoiles Montantes Project "SymFol" funded by Région Pays de la Loire**, 114700€.

03/2023 **Funding PEPS "Jeunes chercheurs et jeunes chercheuses" by INSMI**, 4000€.

2015–2018 **Fellowship "Chapeau Monge"**, École polytechnique.

2015–2018 **PhD grant "Allocation spécifique pour normaliens"**, École Normale Supérieure.
2012–2015 **Master scholarship**, École Normale Supérieure.
2009–2012 **Bachelor scholarship**, Scuola normale superiore.

Languages

Italian (native), English (advanced), French (advanced), German (A2 level)

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